

Review-II — what to say, slide by slide

GaN Based Synchronous Buck Converter with an Improved Gate Driver

Sanjay Kumar 23BEC1447 · Aamir Abdullah 23BPS1197 · Amritha S 23BEC1368

Guide: Dr. Bindu — SENSE, VIT Chennai · Review-II, 30.09.2026

25 slides. **Total 14 min 25 s of speaking.** Bracketed times are cumulative:

if the clock is past one, you are behind. Everything in **bold** is a number

you say out loud, and every one of them is on the slide behind you.

Slides	Who
1–7	Amritha — problem, aim, architecture, mechanism
8–16	Sanjay — schematics, method, validation, the two runs
17–25	Aamir — live demo, results, cost, conclusions

The four that decide the review: 4, 12, 20 and 22 — the architecture, the three device models, the loop-inductance result, and the conclusions. If you run short, cut 9, 10 and 13. Never cut those four.

Do not read the captions aloud. They are there so the panel can check you afterwards. Say the lines below and point at the picture.

1 · Title — 25 s (0:25)

Get the signed slide up first. Do not read the registration numbers.

Good morning. Our project is a **GaN** synchronous buck converter, and the part we designed is the gate driver inside it. It takes **100 V** in and gives **48.50 V** out.

One thing before any result: the size of the fault we study depends on the device model. On one of our three models the off-state gate never crosses threshold at all. So we do not claim the fault is universal — we claim our fix holds on every model we tried.

2 · Problem Statement & Background — 55 s (1:20)

A half-bridge is two transistors in series across the supply. Exactly one is on at a time. If both are ever on, the supply is shorted through them.

GaN is used because it switches fast. That is the reason to choose it, and it is also what causes our problem. When the bottom device turns on, the midpoint collapses in about three nanoseconds. A small capacitance inside the top device, between drain and gate, turns that fast change into a current — and it goes into the gate that is meant to stay off.

We measured it. **1.65 V**. The device turns on at **1.4 V**.

Fixes exist in the literature. But they report their gains as one number, and that number mixes two different things: picking one good setting, and changing the setting while the converter runs. Only the second needs a sensor, a lookup table and a controller. That is our gap.

3 · Aim and Approach — 40 s (2:00)

So our aim is one question: how much benefit does runtime adaptation add beyond a fixed setting, and under what conditions is it worth the hardware?

Four steps. Build the converter and check it converts. Recreate the fault. Fix it one change at a time, each measured on its own run. Then ask whether one fixed setting is enough.

Two tools. **ngspice** runs every simulation. **KiCad** draws every schematic, from the same netlist ngspice runs — so the picture and the simulation cannot drift apart.

4 · System Architecture — 45 s (2:45)

Point along it. Do not read the boxes.

A PWM command comes in. The FPGA holds the control word — six fields, which is **720** possible words. The segmented driver shapes the switching edge. The half-bridge drives the filter and the load.

The green block is what this project designs, and this is what is inside it: eight pull-up segments, eight pull-down, an always-on Miller clamp, and

a mux that selects the off rail at zero or minus two volts.

Two things to notice. The word is set at power-up and never changed — no sensor, no lookup table in this path. And the red arrow is the fault: the switching edge coupling back into the gate that should stay off.

5 · Novelty: Three Added Blocks — 40 s (3:25)

The base paper is Zhang and others, ISPSD 2020. Only the shaded rows differ.

The command is theirs. The power stage is theirs. Segmenting the output stage is theirs — we do not claim it.

Three blocks are added: a digital control word where they use one analogue bias resistor, an always-on Miller clamp they do not have, and an off rail selectable to minus two volts where theirs is tied to zero.

At our headline corner that takes the crosstalk margin from **0.407 V** to **2.576 V**, from one fixed setting.

6 · Crosstalk Mechanism at Device Level — 50 s (4:15)

Slow down — this is where you show you understand the physics.

Left, the mechanism. The bottom device turns on, the switch node falls from **106 V**, and at its steepest that edge is **124 V/ns**. That rate times the drain-gate capacitance is a current with nowhere to go but the top gate.

Right, what the driver does about it. Three paths hold that gate down: the pull-down segments, the Miller clamp with its own half-ohm path timed to the other device's edge, and the off rail that sets where the gate starts.

The important point is that the clamp and the rail are not the same fix. The rail moves where the gate starts; the clamp shortens how far it gets lifted. Two mechanisms, so they add rather than overlap.

7–9 · The three schematics — 30 s total (4:45)

One sentence each. Do not dwell — they are there to be looked at.

(7) The converter as an actual KiCad sheet, drawn from the netlist, with the two gate drivers as sub-sheets of this page.

(8) Our driver: eight pull-up segments, eight pull-down, the Miller clamp on the right, the off rail selectable to minus two volts.

(9) The base paper's driver, rebuilt by us in the same testbench: seven segments per bank in two stages, one bias resistor, no clamp.

10 · Segmented Driver: SPICE Implementation — 25 s (5:10)

This is the actual source of that output stage, at its real line numbers.

Each segment is one switch and one resistor: in circuit when the control word reaches it, one gigaohm when it does not. The clamp is deliberately not one of the segments — it has its own switch and its own timing, because it must hold the gate down while the *other* device is switching.

11 · Methodology — 40 s (5:50)

If you get one methodology question, it is this slide.

Every comparison in this deck is the same file with one thing changed.

Held identical: the netlist, byte for byte — the same file, not a copy. The device model. The parasitics. The operating point. The solver settings.

What changes is named in the right column. For theirs against ours, only the driver subcircuit is swapped.

And we give their driver an advantage its own paper does not have: we search both of its controls at every corner and run it at whatever wins, while ours runs one fixed word everywhere. Every choice here runs against

US.

12 · Model Validation: Three Device Models — 50 s (6:40)

This answers "why should I trust your simulation". Point at the last column, not the first.

Three independent device models: our behavioural model, real SKY130 transistors, and the same model with a charge-based capacitance.

Read the first column. Minus **0.249**, minus **0.563**, plus **0.115**.

The sign changes. On the third model the gate never reaches threshold, so the fault does not occur at all. We are telling you that, because it is true and it bounds what we claim.

Now the last column — the configuration we ship. Plus **2.576**, plus **2.032**, plus **2.710**. Positive on every model, same ordering on every model.

So the existence of the fault is model-dependent. The sufficiency of the fix is not. The second is our claim.

13 · Numerical Reliability — 40 s (7:20)

Two questions this answers: are these numbers physics or solver settings, and where is every run.

Left, total switching energy against timestep, over a twenty-five times range. Right, every metric as a percentage of its own converged value. Flat is the result: the margin moves **0.14** percent, the gate peak **0.04** percent, and no feasibility verdict changes.

On the grid: **25,911** completed runs of 25,920. Nine did not, and we went and found out why. All nine fail again on a re-run, all nine are the same ngspice convergence failure at the high-side gate, and all nine are half-fixed settings — clamp without the rail, or rail without the clamp.

None is the configuration we ship.

14–15 · The two runs — 60 s total (8:20)

These carry video. Click, let it run, talk over it.

(14) Their driver in our testbench. The gate that should stay off peaks at **0.993 V** against a **1.400 V** threshold — **0.407 V** of margin.

(15) The same sequence, our driver. It peaks at minus **1.176 V** — **2.576 V** of margin, six times theirs. Same axes, fixed in the script, so the two pictures are directly comparable. And this is one fixed control word, the same one used at every corner in the study.

16 · Live Simulation — 60 s (9:20)

Run it. Terminal open in the repository beforehand, font already large.

I will run it now rather than show a recording.

```
bash proof/LIVE-SIM.sh
```

While it runs, about ten seconds:

Two transients of one circuit file. Between them, exactly two things change: the Miller clamp is enabled, and the gate off rail is moved.

When the numbers print:

It prints what it just measured next to what our slides claim. The off gate reaches **1.649 V** against the **1.65** on our slide, and the margin is **2.576 V**. Then it draws the two waveforms.

If it fails, say this and move on — do not debug in front of the panel:

The recorded run is on the slide behind me and the script is in the repository.

If asked to see the converter actually converting — the input and output voltages, not just the gate — there is a second command, about twenty-three seconds:

```
bash proof/LIVE-BUCK.sh
```

This runs the whole converter. It prints the output voltage, power and efficiency it just measured beside what our slides claim — **48.50 V**, **236.26 W**, **97.49 %** — and then draws four plots: the output settling to 48.50 V with 729 mV of ripple, the switch node against the 100 V bus, the gate drive showing the -2 V off rail, and the inductor current.

That picture is the answer to "show me 100 volts in and 48.5 volts out". It is not a slide on purpose — it is better drawn in front of them than printed.

17 · Crosstalk: Fault and Mitigation — 45 s (10:05)

Point at the four panels in order.

Two runs, four panels. Top row the cause, bottom row the effect.

Top: the switch node falls, at its steepest **124 V/ns**. Identical in both runs.

Bottom left, no clamp and a zero-volt off rail: the gate starts near zero, is lifted **1.643 V**, peaks at **1.649 V** — past the **1.4 V** threshold.

Bottom right, same run with the clamp on and the rail at minus two volts: it starts at minus **1.941 V**, is lifted only **0.765 V**, peaks at minus **1.176 V**.

You can read both fixes off this one picture. The rail moved the start; the clamp cut the lift from **1.643** to **0.765**.

18 · Comparison with the Base Paper — 55 s (11:00)

The slide the review turns on.

Four corners, mildest to hottest. Only the driver changes.

Two things matter here, and neither is the ratio.

First, the gap widens under stress. Their best margin falls **0.503** to **0.181 V**; ours goes **2.757** to **2.251 V**. They degrade exactly where it matters, because a clamp does not care how hot the device is.

Second — and this is what we would defend hardest — their scheme buys its margin by slowing the edge down. Ours runs about twice as fast and still wins by a factor of several. The margin is not paid for with switching speed.

And the bottom line: their netlist is not published, so every ratio here is against our implementation of their scheme, not their measured result.

19 · Why GaN and Not Silicon — 35 s (11:35)

Before the driver comparison, the question underneath the whole project: why GaN at all.

Same converter, same netlist, same operating point. Only the device and its rated gate drive change — and we matched the on-resistance, **25.0** against **24.0** milliohms, so conduction loss is equal by construction. What is left is switching, gate drive and reverse recovery.

GaN wastes **6.22 W** against silicon's **12.63 W** — **51** percent less — and holds **97.42** percent efficiency against **95.06**.

One caveat we state ourselves: the two do not deliver identical output power, because GaN has no body diode and its dead-time reverse drop is larger. That is the same mechanism that makes our minus two volt rail cost turn-on energy. Loss per watt delivered normalises for it: **2.65** against **5.19** percent.

20 · When Runtime Adaptation Is Worth Building — 50 s (12:25)

~~Say this as a finding, not as an apology.~~

Our headline says adaptation adds two-point-six percent. That is conditional, and this is the condition.

X-axis, power-loop inductance — which is board layout, not the transistor.
Y-axis, the ceiling on what any scheduling scheme could return. At one-point-five nanohenries it is **13.5** percent. At four-point-five it is **0.55** percent.

So "is adaptive gate control worth a sensor, a lookup table and a controller" is not a property of the driver at all. It is a property of the board it sits on. Below about two-and-a-half nanohenries, yes. Above, no.

Two honesties: the series is not monotonic in that band, which is why we plot eight points and not a curve — and three nanohenries is a nominal condition, not a measured layout.

If she asks why the plot reads 6 % at 3 nH when you claimed 2.6 % —

this is the one trap on this slide, and the answer is three quantities:

This axis is the ceiling on scheduling over four corners, which is **5.95** percent at three nanohenries. Over the full thirty-six corner grid that ceiling is **3.5** percent. And what adapting per operating point actually returns is **2.6** percent. The ceiling is what a perfect scheduler could reach; two-point-six is what one delivers.

21 · The Cost of the Fix — 35 s (13:00)

What the margin is paid for with.

It buys **2.825 V** of gate margin. It costs **0.24** points of efficiency, **0.570 W** of loss, and twelve volts of extra switch-node peak.

About **0.22 W** of that loss is third-quadrant conduction — GaN has no body diode, so in dead time it conducts in reverse and pays the threshold plus whatever off-bias we applied. The rail that buys us margin is paid for right there.

22 · Conclusions — 45 s (13:45)

Six, quickly.

One: the clamp with a negative off rail raises the gate margin on all three models. Two: a good fixed word is worth **26.5** percent; re-tuning while running adds **2.6** percent more. Three, and the one we would defend hardest: that answer is conditional — board layout decides it, not the driver. Four: it is a trade, and slide twenty-one is the bill. Five: simulation only, nothing built. Six: a measured waveform that does not match ours, or a loop inductance below two-and-a-half nanohenries, would change conclusion three.

23 · Next Steps — 25 s (14:10)

What remains is bench work, not more simulation. Place-and-route on a chosen board — synthesis is done, the pin constraints are placeholders. And a hardware half-bridge, measured. That is the whole of the remaining risk.

24 · References and close — 15 s (14:25)

These are the papers this design engages with; number ten is the base paper, the one we reimplemented and measured against.

Everything in this deck runs on request from the repository. Thank you.

If she asks

"What is actually your contribution?"

Three blocks the base paper does not have — a digital control word, an always-on clamp, a selectable negative off rail — and a measured answer to a question the field reports as one number: what re-tuning is worth alone.

"And the answer?"

A better fixed word is worth **26.5 %**. Adapting per operating point adds **2.6 %** — about **8.9 %** of the gain. The ceiling on any scheduling scheme is **3.5 %**. So the sensor and the lookup table buy the small share.

"Why should I believe the numbers?"

Every one regenerates from a named script, and a checker fails the build if a slide and the data disagree. The command on slide 16 re-derives two of them in ten seconds.

"Is the clamp your idea?"

No, and the slide says so. Clamps and negative off-bias are established practice; we cite four papers. What is ours is that the base paper has neither, and that we measure what each is worth here.

"Have you built it?"

No. Simulated in ngspice on one behavioural GaN model written from the EPC2010C datasheet. That is the largest limitation, it is on slide 23, and a measured half-bridge is the next step.

"Why is your turn-on energy worse?"

GaN has no body diode. In dead time the device conducts in reverse and pays the threshold plus the off-bias. The rail that buys crosstalk margin is paid for there — we measure it at 1 to 3.4 % of total loss.

"What did you synthesise the controller with?"

Vivado, targeting an xc7a35t — **20 LUTs** and 20 flip-flops. It is synthesised and verified, not placed and routed; the pin constraints are placeholders, which is on the next-steps slide.

"Why ngspice and KiCad?"

Both are open source, so nothing here needs a licence to reproduce. KiCad draws every schematic from the same netlist ngspice runs, so the picture and the simulation cannot drift apart.

Before you walk in

☐ Slide 1 signed by Dr. Bindu and scanned back in as slide 1.

☐ `bash proof/PREFLIGHT.sh` on the presentation laptop — it rehearses both live commands and prints READY.

☐ Terminal open in the repository, font size up, slide 16 rehearsed once.

☐ Videos on slides 14, 15 and 16 play — open the .pptx, not the PDF.

☐ The PDF on a pen drive as the fallback.
